

ZMap: Fast Internet-wide Scanning and its Security Applications

Problem Statement

The paper presents the challenge of conducting Internet-wide network scans efficiently. Traditional tools like Nmap require significant time and resources, making large-scale network analysis difficult for researchers while being more accessible to attackers using botnets. The authors propose ZMap, a high-speed network scanner capable of surveying the entire IPv4 address space in under 45 minutes using a single machine. The paper discusses its architecture, performance, and security implications.

Assumptions

The author make a couple of basic assumptions.

- High-speed scanning is achievable
- Minimal packet loss is acceptable
- IPs of machines which are doing the scan are not blacklisted due to their behaviour.

Pros

I basically like the idea upon which Z-map is developed which results in various advantages.

- Can scan the entire IPv4 space in under 45 minutes, over 1300× faster than Nmap.
- Lightweight & Open Source hence easily customizable.
- Runs entirely in user space without requiring kernel modifications.
- Provides high accuracy (~98% network coverage) even with a single probe.
- Avoids issues caused by traditional scanning tools' per-connection state tracking. Security Applications.
- Good architectural choice of modular design, allowing for different probe types (TCP SYN, ICMP, UDP).

Cons

The immediate Con i thought was that too much packets can make us be detected in certain monitoring systems, hence:

- Even though ZMap randomizes scanning, high-speed probing can overload networks. Some system administrators may misinterpret scans as malicious attacks. Ethical & Legal Concerns
- No explicit consent from scanned networks—some may object to unsolicited probes. Researchers must adopt good scanning practices.
- Limited IPv6 Support
- No Packet Retransmission
- Unlike Nmap, ZMap does not retry failed probes, leading to some missed hosts (~2%).

I am not counting the way malicious people can use this tool as a con of this tool, if someone uses a knife badly it does not mean that is a bad point for the knife.